



Microbial diversity analysis of Chumathang geothermal spring, Ladakh, India

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Abstract

In light of their unique and challenging environment, the high-altitude Chumathang geothermal springs in Ladakh, India, are undeniably intriguing for microbiological study. The purpose of this study was to employ a culture-independent sequencing approach to give a comprehensive characterization of the unknown bacterial and archaeal community structure, composition and networks in water and soil from the Chumathang geothermal spring. A total of 50%, and 42.86% bacterial phyla were found in the water, and soil samples respectively and this analysis also showed a total of 9.62% and 7.94% of archaeal phyla in both the samples, respectively. Further, the presence of unclassified (derived from other sequences, water: 17.31%, and soil: 19.05%) and unclassified (derived from bacteria, water: 13.46%, and soil: 12.70%) were also observed in the current metagenomics investigation. Firmicutes and Proteobacteria were the most abundant bacterial phyla in water, whereas Proteobacteria and Bacteroidetes were the most abundant bacterial phyla in geothermal soil. Crenarchaeota and Euryarchaeota dominated archaeal communities in soil and water, respectively. This metagenomic study gave a detailed insight into the microbial diversity found in Chumathang geothermal spring and surrounding area, located in Ladakh, India. Surprisingly, this finding indicated the existence of geographically distinct microbial communities that were suited to various geothermal water habitats along the Himalayan Geothermal Belt. Future studies must take into account the metabolic pathways of these microbial communities that exist in these extreme environments. This will allow us to obtain a better knowledge of the microbial metabolisms that are common at these geothermal locations, which have a lot of potential for biotechnological applications. They will also enable us to establish links between the microbial community composition and the physico-chemical environment of geothermal water and area.

Keywords Geomicrobiology · Extremophiles · Bacteria · Archaea · Metagenomics studies · Functional analysis

Introduction

Microorganisms are the key component of all environments. Because of the ease of dispersal, small size, ability to assimilate a broad spectrum of nutrients, and ability to tolerate extreme environmental conditions, these organisms may colonize domains that are unsuitable for many living

forms [1]. Extremophiles are microorganisms that thrive in extreme environments (e.g., cold, hot, acidic, and alkaline environments) and are further classified as psychrophiles, thermophiles, acidophiles, and alkaliphiles depending on their environmental inclinations [2]. Extremophiles can develop specific physiological characteristics as an adaptive mechanism due to the rare and distinctive geophysical and physicochemical aspects of these extreme environments [3]. Investigating microbial communities that thrive in harsh environments is thus expected to find novel and unique species that can be employed as biologically active substance sources for therapeutic and commercial uses [4, 5]. Extremophilic microorganisms are also fascinating because they can be employed in studying the origins and evolution of life [3].

Extreme habitats like geothermally heated springs are examples where unique and fascinating extremophiles can be found [6]. Geochemistry, pH and temperature of

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the water, according to geothermal spring metagenomics research, influence the most prevalent phylum and taxa within each phylum in diverse geothermal springs. More in-depth investigation is required to comprehend the complete microbial diversity of geothermal springs. Along with the unusually high temperature, geothermal water can typically be distinguished from non-geothermal groundwater by several dissolved minerals, including sodium, calcium, magnesium, chlorides, boron, sulphates, or silica [7, 8]. For several decades, researchers have studied the microbial composition of thermal ecosystems on various continents around the world, including geothermal springs on the Tibetan Plateau of China [9], silica-deposited Yellowstone National Park thermal springs in USA [10], acidic thermal pools in Russia [11], intertidal geothermal springs in Iceland [12], and alkaline geothermal springs in Kenya [13], utilising both traditional culture-dependent methods [14] and novel approaches for high-throughput sequencing [13]. In geothermal springs, novel species capable of yielding valuable biotechnological products such as bioethanol [15], antibiotics [6, 15], and thermostable enzymes [16] have frequently been identified. In India, the geothermal springs of Balarampur, Bakreshwar, Panamic, Chumathang, Vashisht, and Manikaran have been identified as unique sites that may be rich in novel genes and microbes with distinctive attributes [17].

Ladakh is a Trans-Himalayan cold desert in the Indian subcontinent that is 45 million years old, 9000 to 25170 feet high, and has temperatures between +30 and -35°C [18, 19]. The microbes in this biosphere have a special ability to endure and adapt to adverse environmental circumstances, such as insufficient nutrients, extremely low temperatures throughout the year (chilling stress), and brief, intense warmth with UV radiation during the summer [20]. However, there are only a few studies on microbial bioactive from hostile niches like Ladakh. These include alkali-tolerant dye-degrading *Aspergillus* sp. strain CB-TKL-1 [21], *Bacillus lehensis* strain MLB2 for the production of alkaline protease [22], and *Pseudomonas* sp. ICTB-745 that produces antimicrobial metabolites like 1-hydroxyphenazine, phenazine-1-carboxylic acid, rhamnolipid-1, and rhamnolipid-2 [23]. The Chumathang geothermal area (10442 ft, 33°17' N; 78°21' E, [24]) is located in the Indus Valley in the North-West Himalaya's eastern Ladakh region. This area has geothermal springs that provide thermal activity [25]. The thermal discharges (85 to 87°C) at Chumathang have an alkaline pH and are made of sodium bicarbonate-chloride water [25]. Studies on oxygen and hydrogen isotopes show that the Chumathang hot waters are mostly meteoric in origin [25]. Traditional healers (Alchi and Amchi) in Ladakh have used geothermal springs as a treatment option for people suffering from joint pain. Hence

this resource has been used both directly and indirectly for bathing, washing, cooking, and medicine from the dawn of humanity [26]. Because of these factors, it is crucial to look for novel bioactive metabolites produced by microorganisms isolated from Chumathang geothermal springs in Ladakh, India.

Thus, keeping in view the potential of microbial communities in various industries, the focus of this study was to present a thorough assessment of the structure, composition, and networks of microbial communities inhabiting alkaline water and sediments from the underexplored high-altitude Chumathang geothermal springs. The Illumina platform was utilized for whole genome metagenomics sequencing to investigate the microbiological composition of these environments, which had been anticipated to contain unique microbes adapted to extremely challenging conditions in the environment. The findings of this study strengthen the knowledge of microbial networks in geothermal spring ecosystems and improve our understanding of the microbial biodiversity in the Chumathang geothermal springs.

Material and methods

Site description and sample collection

Three water (CuW), and three soil (CuSo) samples have been collected from the Chumathang geothermal spring and its nearby area in the Himalayan Geothermal Belt (Trans-Himalayan Plateau), Ladakh, India on September 10, 2021 (Fig. 1). All samples were collected in the sterile thermostable sample containers, stored in iceboxes, and later transferred to 4°C until further analysis. The physiological properties of the samples recorded in situ are briefed in Table 1.

Isolation and quantification of environmental DNA

Water (03), and soil (03) samples were pooled, and the pooled samples of water (01), and soil (01) were used for the metagenomic study. Environmental DNA was recovered from water samples using the DNeasy PowerWater Kit (Qiagen, Hilden, Germany), while sediment and soil samples were isolated using the DNeasy® PowerSoil® kit (Qiagen, Hilden, Germany). The amount of isolated DNA was quantified using a Nanodrop (ThermoFisher Scientific, Massachusetts, USA) and gel electrophoresis. Final libraries were measured using a Qubit 4.0 fluorometer (ThermoFisher Scientific, Massachusetts, USA, #Q33238) and a DNA HS assay kit (ThermoFisher Scientific, Massachusetts, USA, #Q32851) following the instructions provided by the manufacturer.

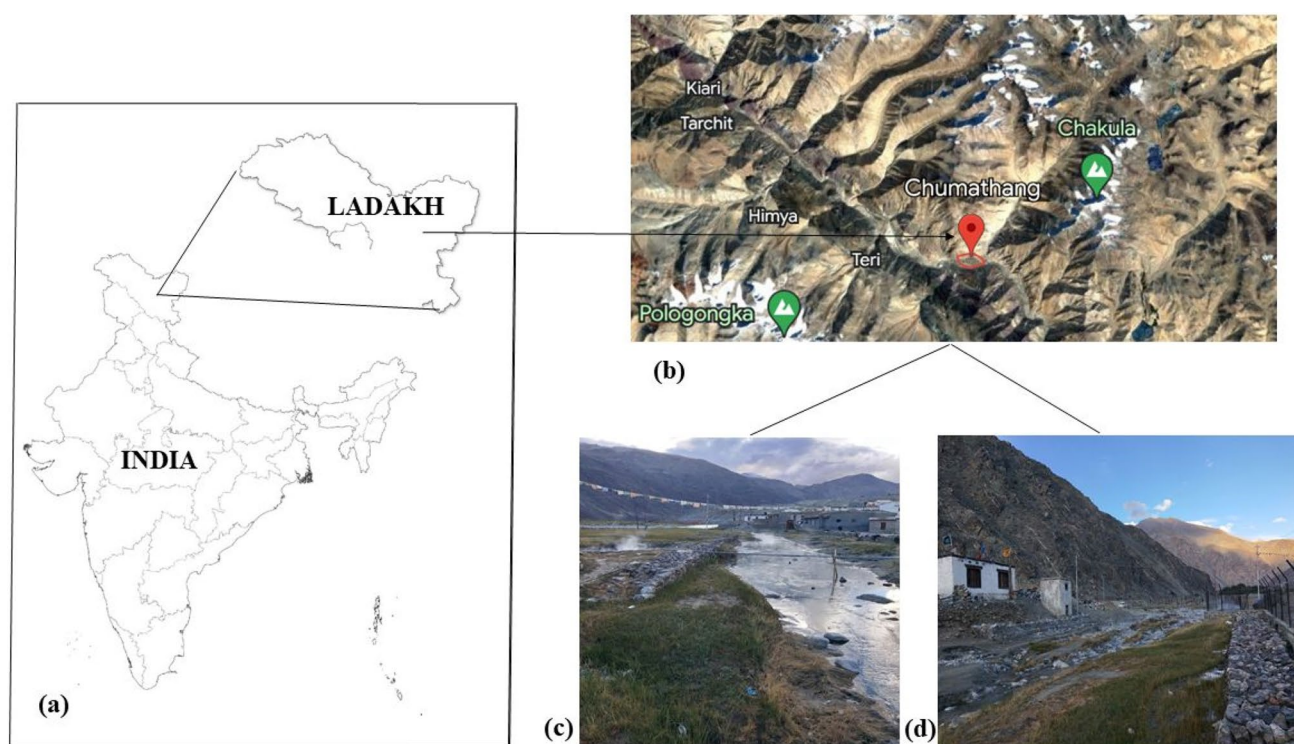


Fig. 1 Location sites (a) schematic map showing the location of the study site in the Western Indian Himalayan Geothermal Belt (Ladakh, 32°N to 36°N and 75°E to 80°E) region, (b) Satellite view of the Chumathang geothermal spring area, sourced from the publicly avail-

able database, <https://www.google.co.in/maps>. (c) and (d) both represented Chumathang geothermal spring and nearby areas (33°17' N and 78°21' E)

Table 1 Sample quality control and sequence statistics of raw and trimmed data

Attributes	CuW	CuSo
No. of samples	1	1
Raw Data		
Total reads	51,097,990	72,191,530
Total bases	8,124,580,410	11,478,453,270
Total GB	8.12458041	11.4784533
Average GC (%)	47	57
Length (bp)	159	159
Total sequences (Millions)	25.5	36.1
Trimmed Reads		
Total reads	50,385,040	68,082,226
Total bases	7,987,403,959	10,006,348,522
Total GB	7.98740396	10.0063485
Average GC (%)	47	58
Length (bp)	149	135
Total sequences (Millions)	24.8	35.1
Bacterial phyla (%)	50	42.86
Archaeal phyla (%)	9.62	7.94
Unclassified (derived from other sequences, %)	17.31	19.05
Unclassified (derived from bacteria, %)	13.46	12.70

Whole-genome shotgun metagenome sequencing and data analysis

The library was sequenced using the Illumina NovaSeq 6000 platform (Illumina, USA). The insert size of the library was determined using the TapeStation 4150 (Agilent Technologies, Palo Alto, CA, USA) and high sensitive D1000 screentapes (Agilent # 5067- 5582) in accordance with the manufacturer's methodology. FastQC v.0.11.9 was used for assessing the quality of the raw fastq reads from the sample [27]. The raw fastq reads have been processed using Fastp v.0.20.1 [28], then quality checked with FastQC and summarised with MultiQC [29]. The processed reads were submitted to taxonomic assignment using MG-RAST [30] against the Silva SSU Databases at six taxonomic lineage levels: phylum, class, order, family, genus, and species. The processed readings were then subjected to Taxonomic Classification using KRAKEN2 [31]. The OTU file in BIOM format was generated from the KRAKEN2 output using the tool Kraken-boom (<https://github.com/smdabdoub/kraken-biom>). KronaTools (<https://github.com/marbl/Krona>) was used to generate Krona Plots of the individual samples. The merged OTU table in biom format was submitted to AMPVIS2 to generate the OTU Clustering Heatmaps, PCA plot, and Alpha Diversity plots [32]. Post Analysis Queries: Based on the top 50 average abundance of the OTUs, phylogenetics was constructed using the NCBI common tree [33] and visualised using Interactive Tree Of Life (iTOL; [34]). The data are publicly available in NCBI Sequence Read Archives (SRA) Accession Nos: SRR24762751 (soil) and SRR24762752 (water) with BioProject ID. No.: PRJNA977415 and BioSample ID. No.: SAMN35523826.

Results

Physiological parameters of sampling site

The physiological properties of the Chumathang geothermal spring are presented in Supplementary Table 1. The temperature at the geyser ranged between 25 ± 2 °C to 87 ± 2 °C, with pH values of 8.6. The total dissolved solids (measured in ppm) ranged from 320 to 625. These measurements were all made on-site, in real-time.

Metagenomic analysis

A total of 06 samples i.e. water (03), and soil (03) were collected from the Chumathang geothermal spring, Ladakh and, pooled samples of water (01), and soil (01) were used for DNA sequencing analysis to recognize novel microorganisms. These samples were all obtained in the vicinity of the Chumathang geothermal spring in Ladakh, India.

These samples were prepared by creating cDNA libraries for Illumina sequencing, and the sequence data obtained from this technique was examined using informatics pipelines (MG-RAST and KRAKEN2) designed to detect known or novel microorganisms (Table 1). KRAKEN2 was used for taxonomic classification and MG-RAST was used for taxonomic, functional analysis, alpha diversity and, phylogenetic analysis. As raw data, the MG-RAST and KRAKEN2 systems yielded 51,097,990 and 72,191,530 reads, respectively, with an average length of 159 bp in both the samples, namely water (CuW) and soil (CuSo). The metagenomic data were put together using Newbler, a program that makes algorithmic attempts to join shorter contigs together from individual sequence reads. For bioinformatics analysis, a total of 50,385,040 (CuW), and 68,082,226 (CuSo) readings that passed MG-RAST quality filters were employed (Table 2). The trimmed read lengths (in bp) were 149 (CuW) and 135 (CuSo) and the average GC (guanine-cytosine) content across all contigs were 47% (CuW), and 58% (CuSo).

Microbial taxonomic diversity

The taxonomic classification of high throughput metagenomic sequencing data for the Chumathang geothermal spring sample processed with the KRAKEN2 database showed the domain bacteria as the most abundant domain (CuW: 95.47%, and CuSo: 90.45%, Supplementary Fig. 1). MG-RAST analysis of whole genome sequencing showed a total of 50%, and 42.86% bacterial phyla in the CuW, and CuSo samples respectively (Table 1) and this analysis also showed a total of 9.62% and 7.94% of archaeal phyla in both the samples, respectively. Further, the presence of unclassified (derived from other sequences, CuW: 17.31%, and CuSo: 19.05%) and unclassified (derived from bacteria, CuW: 13.46%, and CuSo: 12.70%) were also detected in the current metagenomics study. In the CuW sample, Firmicutes (55.65%) has been shown to be the most prevalent phylum (Fig. 2) and the second most abundant phylum was Proteobacteria (15.51%) followed by Aquificae (7.46%), Bacteroidetes. (5.23%), Deinococcus Thermus (5.23%). Proteobacteria (51.31%) was most abundantly present in the soil sample i.e. CuSo. Bacteroidetes phylum ranked second in abundance in CuSo with 16.63%, followed by Actinobacteria (11.44%), Euryarchaeota (9.81%), Planctomycetes (2.40%), and Deinococcus Thermus (1.30%). Apart from these bacterial phyla, five archaeal phyla namely, Crenarchaeota, Euryarchaeota, Korarchaeota, Nanoarchaeota, Thaumarchaeota were also identified in both samples. Crenarchaeota was found to be most dominant in CuW with an abundance of 1.36% and in CuSo, Euryarchaeota (9.81) was dominant archaeal phylum. Further, 99.21% sequences were bacterial, 0.43% were eukaryotic, and 0.11% were archaeal in origin, according to MG-RAST-assisted community analysis. At the

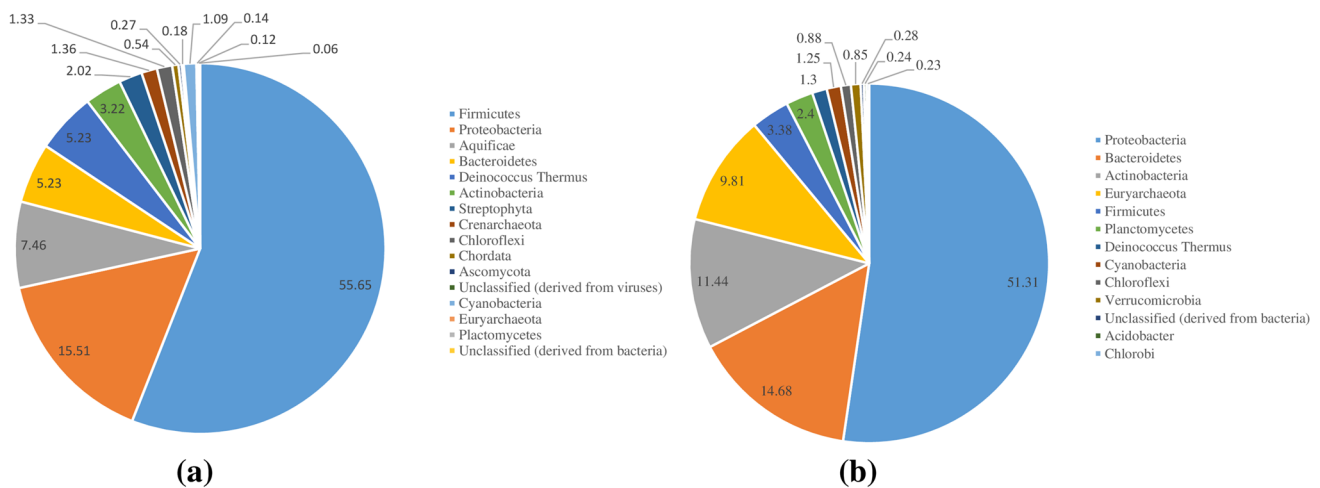


Fig. 2 Pie chart showing the taxonomic distribution of the samples (a) CuW, and (b) CuSo at the Phylum level

genus level, variations in abundance have also been noted between the CuW and CuSo sample of Chumathaang (Supplementary Fig. 2). In CuW sample, most dominant genus was *Bacillus* (31.21%) followed by, *Paenibacillus* (8.43%), *Acinetobacter* (7.49%), *Thermus* (7.16%), *Pseudomonas* (5.64%), *Algoriphagus* (5.08%), *Hydrogenobacter* (2.46%), *Propionibacterium* (2.07%), *Lactobacillus* (1.90%), *Planococcus* (1.70%) and *Lactococcus* (1.55%). However, in CuSo sample, *Algoriphagus* (5.02%) was revealed as the dominant genus and the second abundant genus was *Halalkalicoccus* (3.57%), followed by *Haloterrigena* (2.55%), *Natrialba* (2.06%), *Marivirga* (2.02%), *Rhodobacter* (1.92%), *Pseudomonas* (1.77%), *Acinetobacter* (1.70%), *Bacillus* (1.62%), *Streptomyces* (1.60%), *Mesorhizobium* (1.31%) and *Halomonas* (1.16%). In case of archaeal genera, the most prevalent genus was *Halalkalicoccus* (3.57%) followed by *Haloterrigena* (2.55%), *Natrialba* (2.06%) in CuSo sample whereas in CuW sample, *Aeropyrum* (0.22%) was the predominant genera followed by *Acidilobus* (0.11%).

In the present study, a total of 1953 number of species in CuW sample and 2520 number of species in CuSo sample were reported. The most abundant species in CuW sample was *Bacillus* sp. NRRL B 14911 (6.32%), followed by *Acinetobacter baumannii* (3.35%), *Bacillus subtilis* (2.88%), *Aquifex aeolicus* (2.18%), *Thermus thermophilus* (2.08%), *Bacillus cereus* (1.94%), and *Hydrogenobacter thermophilus* (1.92%) etc., and *Rhodobacter sphaeroides* (2.09%), and *Algoriphagus* sp. PR1 (1.6%) are the abundant species found in CuSo sample (Fig. 3 and Supplementary Fig. 3). Similarly, in archaeal diversity, *Haloterrigena turkmenica* (1.37%) followed by *Natrialba magadii* (1.12%), *Halalkalicoccus jeotgali* (0.94%), *Haloarcula marismortui* (0.9%) and *Natronomonas pharaonis* (0.8%) in CuSo sample and in CuW sample, the predominant species is *Aeropyrum* sp. (0.22%) followed by *Acidilobus saccharovorans*

(0.11%), *Pyrobaculum aerophilum* (0.10%), *Hyperthermus butylicus* (0.08%) and *Pyrobaculum arsematicum* (0.07%), etc. Since MG-RAST is inefficient for analysis of eukaryotic and viral sequences, the primary focus of the current study was on bacteria and archaea. Thus, they were not studied in detail. It was further observed that only a small percentage of geothermal springs in the abovementioned geothermal area has been subjected to microbiological research. The data is still very scarce, and thus there is a great scope for research in these extreme environments.

Functional profiling

The metabolic profiles have been generated using COG (clusters of orthologous groups), NOG (nonsupervised orthologous group), KO (KEGG Orthology), and the subsystem database, which compared functional gene homology against the database and exhibited a category of annotated genes with current metagenomic samples. According to the COG and NOG databases, metabolism (CuW: 44.45% and CuSo: 45.67%) and poorly characterised proteins (CuW: 77.32% and CuSo: 78.22%) were the most common protein functional classifications (Supplemental Fig. 4). However, in case of KO database, the metabolism (CuW: 59.79% and CuSo: 60.32%) was followed by genetic information processing (CuW: 16.86% and CuSo: 17.81%) and environmental information processing (CuW: 16.62% and CuSo: 5.91%). Functional classification using a subsystem database (Fig. 4) demonstrated an unambiguous dominance by the Carbohydrates metabolism (CuW: 14.35% and CuSo: 13.81%) followed by Clustering-based subsystems (CuW: 13.91% and CuSo: 13.87%), Amino Acids and Derivatives (CuW: 11.03% and CuSo: 11.44%), Miscellaneous (CuW: 7.42% and

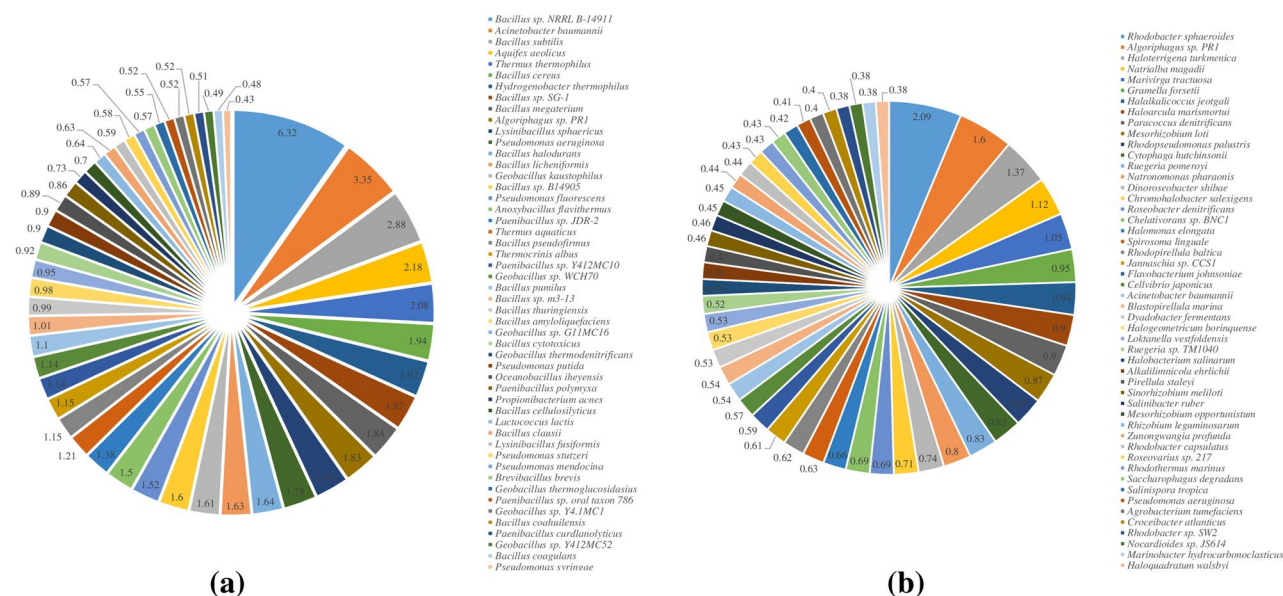


Fig. 3 Pie chart showing the taxonomic distribution of the samples (a) CuW, and (b) CuSo at the Species level

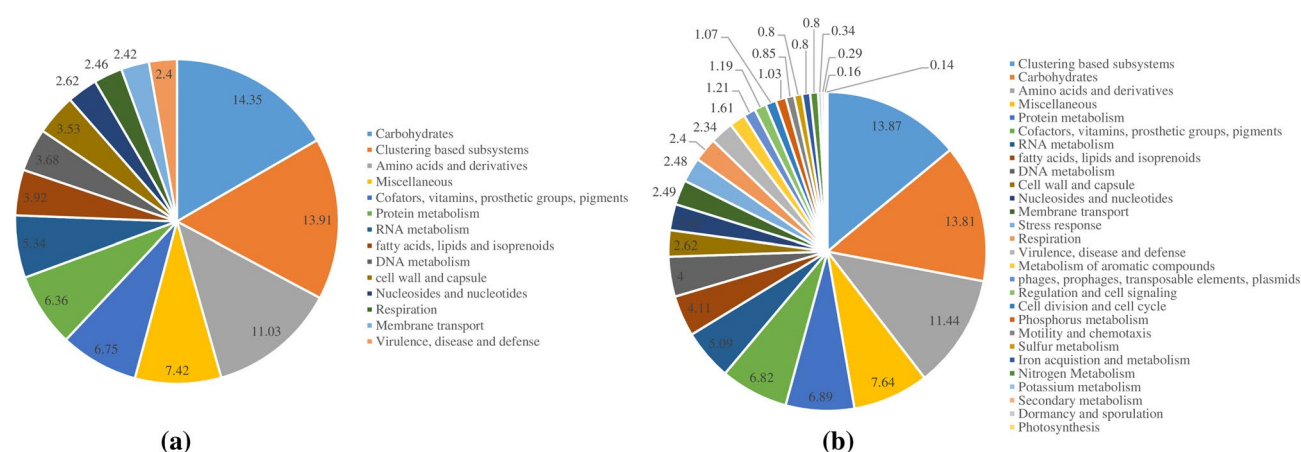


Fig. 4 The pie charts show the distribution of functional groups for the Subsystems for the samples (a) CuW, and (b) CuSo. Each slice indicates the percentage of readings for the given source with predicted protein functions annotated to the category

CuSo: 7.64%), Protein Metabolism (CuW: 6.36% and CuSo: 6.89%), Cofactors, Vitamins, Prosthetic Groups, Pigments (CuW: 6.75% and CuSo: 6.82%), RNA Metabolism (CuW: 5.34% and CuSo: 5.09%), Fatty Acids, Lipids, and Isoprenoids (CuW: 3.92% and CuSo: 4.11%) and membrane transporter (CuW: 2.42% and CuSo: 2.49%) etc.

Phylogenetic analysis

It is used to study the evolutionary relationship between biological organisms. Phylogenetic analysis is crucial for understanding biological diversity, genetic classifications, and the

developmental processes that take place during evolution. The sequence of a gene is currently used in phylogenetic analysis to identify the evolutionary relationships between species as a result of advancements in genetic sequencing technology. The phylogenetic tree and heatmap were based on the most abundant 50 bacterial species in PugW, PugS, and PugSo samples, generated by using NCBI common tree utilizing the top 50 average abundance of the OTUs and visualized the tree using Interactive Tree Of Life (iTOL, [34]). To perform the taxonomy abundance from domain to species level MGRAST bar plots were used. Significant variation among OTUs was observed in all samples, according

to the heat map (Supplementary Fig. 5) and the phylogenetic trees (Supplementary Fig. 6). In both the cases i.e. CuW and CuSo samples, the branching from the domain of bacteria was mostly dense as compare to the domain of archaea and eukaryotes according to the carl woes domain system of classification. The number of OTUs, in case of CuW sample of bacteria were 371219,9 and in CuSo sample, 583710 OUTs were present. As per domain system of classification the phylum Proteobacteria in case of CuSo sample came out with maximum OTUs i.e., 176282. On the other hand, the CuW sample majorally had Firmicutes phylum from rest of the data with OTUs 316439. Evolutionary *Rhodobacter sphaeroides* species from CuSo sample mainly branched from the bacterial domain and came under Proteobacteria phylum that is with maximum OTU counts of 158242. But the dominating strain of CuW sample is *Bacillus* sp. NRRLB-14911 having OTUs 62493 from phylum Firmicutes.

Alpha diversity

A total number of 1954, and 2560 OTUs were detected in samples CuW, and CuSo respectively, Shannon–Wiener Diversity (H), Shannon–Wiener Evenness (EH), and Simpson Dominance (D), Indices were employed to determine and compare diversity of the microorganisms in the samples, based on the metagenomic data of the geothermal spring and its nearby areas. The calculated value of the diversity indices to species evenness and richness were shown in Fig. 5 and Supplementary Table 2.

Discussions

The unexplored diversity of microorganisms in the geothermal springs are believed to be a source of industrially relevant enzymes. As a consequence, the whole genome sequencing metagenomics technique was utilised to characterise the microbial diversity of the Chumathang geothermal spring, as well as to predict the essential functions of the microbiome. The link between various environmental

parameters e.g., temperature and, microbial communities which is considered to be the primary element driving the community structure, has been the subject of numerous investigations of geothermal spring environments [35]. The temperature has no monotonic relationship with microbial diversity, and a range of other spatial or environmental factors may also be involved in determining the microbial community. Other elements that must be considered are available organic carbon, total dissolved solids, and the amount of salt. The physiological properties of the Chumathang geothermal spring are presented in Supplementary Table 1. A geochemical examination of the spring water from geothermal sources revealed high pH levels, high TDS levels, and the presence of trace elements including arsenic and strontium. The chloride component of geologic crustal fluid may be what is causing the enriched chloride ions in the geothermal spring at Chumathang [36, 37]. A relatively low resistance region of 13–30 Ω m in depth up to 300 m was discovered by other magnetotelluric investigations, indicating that there is active geothermal activity at depth [36]. However, low resistivity values in the audio-magnetotelluric data could be the result of modification or the occurrence of clay-rich sediments. Geochemical thermometry projected that the deep reservoir temperature would be at least 145 °C. Regardless, a reservoir temperature of 107 °C was indicated by silica-quartz geothermometry of available data [37].

An essential characteristic of microbial genomes is the average genomic GC content, which has been linked to properties including genome size, exposure to oxygen and nitrogen, and particular environments. In contrast to soil bacteria, which typically have larger genomes and greater percentages of GC, intracellular bacteria often have smaller, more AT-rich genomes [38]. In comparison to the metagenomic analysis of present study (GC contents and reads, Table 1), the Illumina HiSeq yielded sequencing data from 37338796 reads in the Yumthang geothermal spring sample and 61898482 reads in the Reshi geothermal spring sample with a read length of 150. The GC content of Reshi and Yumthang geothermal springs was reported to be 64% and 65%, respectively [39]. Only the high- and near-high-quality MAGs had a GC% that varied from 32.15 to 69.21%, whereas all other MAGs' GC%

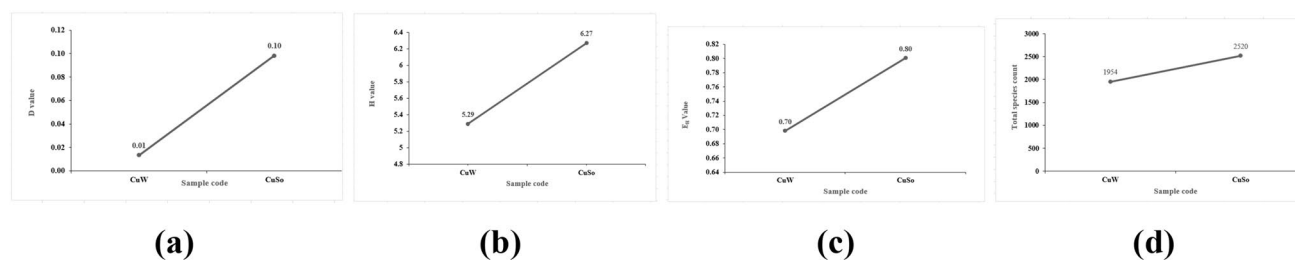


Fig. 5 Simpson Dominance (a), Shannon Diversity (b) and Shannon Equitability (c) Indices, and total species count (d)

ranged from 25.76% to 70.35%, for a metagenomic perspective on the geothermal springs microbiomes in the Subantarctic Kerguelen Islands [40].

In the current study, the metagenomic study displayed abundant bacterial and archaeal communities in water and soil samples of Chumathang geothermal spring. A total of 50% (water), and 42.86% (soil) bacterial phyla and a total of 9.62% (water) and 7.94% (soil) of archaeal phyla along with the presence of unclassified (derived from other sequences, water: 17.31%, and soil: 19.05%) and unclassified (derived from bacteria, water: 13.46%, and soil: 12.70%) were observed in the present metagenomics study. In the water sample, Firmicutes, Proteobacteria, Aquificae, Bacteroidetes, Deinococcus Thermus were present. Proteobacteria (51.31%), Bacteroidetes, Actinobacteria, Planctomycetes, and Deinococcus Thermus were mainly present in the soil sample of Chumathang geothermal spring. Apart from these bacterial phyla, five archaeal phyla namely, Crenarchaeota, Euryarchaeota, Korarchaeota, Nanoarchaeota, Thaumarchaeota were also identified in both samples. The microorganisms present in the soil and water samples of Chumathang geothermal spring were also identified on the basis of genera and species level in the current study. Previous metagenomic investigations from various geothermal environments and geothermal springs revealed a variation in the population distribution pattern of various bacterial phyla. On the metagenomics study for unveiling the wide range of microbes of Lasundra geothermal spring, Gujarat State, India, 29 bacterial, 20 eukaryotic, and four archaeal phyla have been identified [41]. Firmicutes (95.5%) were the most prevalent bacterial phylum in this study, afterwards Proteobacteria (2.0%), and Actinobacteria (0.8%) etc., in this study. The microbial analysis unveiled a dominance of Bacteria ($93.11 \pm 1.86\%$), afterward Archaea ($6.18 \pm 1.84\%$), Eukaryota ($0.67 \pm 0.009\%$), and Viruses ($0.02 \pm 0.03\%$). Furthermore, the most abundant phyla were Proteobacteria ($68.25 \pm 3.59\%$), Aquificae ($11.24 \pm 1.15\%$), and Deinococcus-Thermus ($5.26 \pm 1.01\%$) etc., in bacterial domain out of the 27 bacterial phyla, in a similar study of Burgas geothermal spring [42]. In the metagenome from Lasundra geothermal spring, India (Mangrola et al., 2015), the most predominant genera were *Bacillus* (86.7%), and *Geobacillus* (2.4%) etc. *Thermus*, *Hydrogenobacter*, and *Thiobacillus* were the three abundant genera in the As Burgas water [42], respectively, with 21,221 sequences (15.77%), 11,517 sequences (8.56%), and 5659 sequences (4.20%). In Himalayan geothermal springs, the metagenomics study revealed the abundance of thermophilic (55 °C) bacterial and archaeal taxa present in water as compared to microbial mats e.g., *Methyloversatilis*, *Methylophilus*, *Caulobacter*, *Emticicia*, *Thermus*, *Pyrobaculum* and *Enhydrobacter*. *Pyrobaculum*, *Massilia* and *Chloroflexus* were detected in abundance in the sediment samples [43]. The

archaeal genera *Sulfolobus* and *Metallosphaera* are the most abundant strains in the acidic to circumneutral geothermal springs in the Philippines [44]. The genus *Anaerolinea* was a thermophilic anaerobic genus, members of this genus grow chemo-organotrophically on a broad spectrum of carbohydrates [45]. Thus, the phylum-level geographic distribution of microorganisms (archaea and bacteria) in the current study correlated with prior metagenomic analyses. Overall, the high-altitude geothermal spring water and soil samples from the current metagenomic analysis showed a diverse microbial community comprising heterotrophic, chemoautotrophic, and photoautotrophic microorganisms. Furthermore, the results of metagenomic analysis provided a more credible assessment of the microbial diversity identified in high-altitude geothermal spring water and soil samples.

Further, for the functional analysis of the metagenome of Chumathang geothermal spring, in contrast to present study, COG analysis of the soil from the Aguas Calientes geothermal spring in Peru represented General and Unknown functional categories, with Amino acid transport and metabolism (8.3%) having the highest representation at the second tier, trailed by Energy production and conversion (6.7%) and Transcription (6.6%). According to KEGG analysis, the sequences were classified as Membrane transport (12.1%), Amino acid metabolism (10.3%), and Carbohydrate metabolism (10.2%, [46]). Around 24.4% of the genes in all samples taken from the four geothermal springs in Orisha, India, were associated with carbohydrate and protein metabolism subsystems, whereas 14.6% sequences were associated with nucleic acid metabolism and nucleosides and nucleotides subsystems [47]. Approximately 6.2% and 8.4% of the genes were associated with cofactor, vitamin, prosthetic group, pigment, and amino acid and derivative subsystems, respectively. The survivability of the microorganisms in the geothermal geothermal spring environment may be due to the number of genes associated to various metabolic pathways. For phylogenetic analysis, in both the cases i.e. CuW and CuSo samples, the branching from the domain of bacteria (Firmicutes, Proteobacteria and Bacteroidetes along with Cyanobacteria, and Chloroflexi, etc.) was mostly dense as compare to the domain of archaea and eukaryotes in the present study. Similar research found bacterial clones from four known taxa, notably Proteobacteria, Firmicutes, Chloroflexi, and Cyanobacteria, in the case of geothermal springs on the Tibetan Plateau [48]. According to the phylogenetic analysis, bacteria were more diverse than archaea, and these clone sequences were grouped into 82 bacterial and 41 archaeal OTUs, respectively [48]. The α -Diversity metrics demonstrated more species diversity in the soil samples as compared with water samples. This variation may be due to the abiotic and biotic factors of soil which provide a habitable environment to flourish/support a rich microbial community. The α -diversity (Shanon) differences between

sediment and water samples were connected with various physical–chemical variables including water sample equity, dissolved As_{Tot} , DO, sulphate, temperature, and TOC etc. as also given in the study done by Jiang et al. [49]. Further, the solid As_{Tot} in the samples most likely played a substantial impact in affecting the organization of the microbial community. Although the alpha diversity of all the springs was essentially the same, the beta diversity indicated differences in the degree of uniqueness of OTUs at varied temperatures in Pandanus Rich Thermal Spring Cluster, Deulajhari, India [50].

Conclusion

Finally, a whole metagenome sequencing method was used to investigate the taxonomic and functional characterization of Chumathang geothermal springs in Ladakh, India. The phyla dominated the microbial consortia include Firmicutes, Proteobacteria, Aquificae, Bacteroidetes, Deinococcus Thermus, Actinobacteria, Euryarchaeota, Planctomycetes, Crenarchaeota, Euryarchaeota, Korarchaeota, Nanoarchaeota, Thaumarchaeota. Further, The alpha diversity metrics demonstrated more species diversity in the soil samples as compared with water samples. The Chumathang geothermal spring and its surrounding environment, which are geographically distant, are reservoirs of biodiversity and novel microorganism that should be interesting to investigate in terms of microbial life evolution and speciation mechanisms. It has proven difficult to properly define the microbial metabolic diversity in these geothermal pools due to the inherent restrictions of Metagenome Assembled Genome (MAG) reconstruction and the present state of the comprehension of microbiological pathways. These geothermal environments might, however, be reservoirs of unique biological and genetic material. Large-scale metagenomics, meta-transcriptomics, and cultural analyses should be used in the future to thoroughly explore the physiological characteristics and adaptive mechanisms of microbes living in these distinct habitats.

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Author contributions SD provided significant contributions to the manuscript's conceptualization and drafting. SD and KA collected the samples and data to write the manuscript. SD and KA wrote the manuscript, with support from SK and GC. The submitted manuscript was read and approved by all of the authors.

Data availability This published paper and its supplementary information file [Supplemental Figures and Tables: 2036 kb] comprise all data generated or analysed during this analysis.

Declarations

Competing interests The authors declare that they have no competing interests.

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